





Single subject QC report generated using eddy quad v1.4.2

When using eddy and its QC tools, we ask you to please reference the papers describing the different aspects of the modelling and corrections. The following suggestion for a methods section and list of references has been tailored for you based on your eddy command line.

METHODS
The susceptibility induced off-resonance field was estimated from spin-echo EPI images acquired with different phase-encode directions (Andersson et al., 2003). This field was passed to "eddy", a tool that combined it with estimating gross subject movement and eddy current-induced distortions (Andersson & Sotiropoulos, 2016).The quality of the dataset was assessed using the eddy QC tools (Bastiani et al., 2019). Slices with signal loss caused by subject movement coinciding with the diffusion encoding were detected and replaced by predictions made by a Gaussian Process (Andersson et al., 2016).

REFERENCES
Jesper L.R. Andersson, Stefan Skare and John Ashburner. 2003. How to correct susceptibility distortions in spin-echo echo-planar images: application to diffusion tensor imaging. NeuroImage 20:870-888

Jesper L.R. Andersson and Stamatios N. Sotiropoulos. 2016. An integrated approach to correction for off-resonance effects and subject movement in diffusion MR imaging. NeuroImage 125:1063-1078

Matteo Bastiani, Michiel Cottaar, Sean P. Fitzgibbon, Sana Suri, Fidel Alfaro-Almagro, Stamatios N. Sotiropoulos, Saad Jbabdi and Jesper L.R. Andersson. 2019. Automated quality control for within and between studies diffusion MRI data using a non-parametric framework for movement and distortion correction. NeuroImage 184:801-812

Jesper L.R. Andersson, Mark S. Graham, Eniko Zsoldos and Stamatios N. Sotiropoulos. 2016. Incorporating outlier detection and replacement into a non-parametric framework for movement and distortion correction of diffusion MR images. NeuroImage 141:556-572

Volume-to-volume motion

Average abs. motion (mm)	0.60
Average rel. motion (mm)	0.25
Average x translation (mm)	-0.22
Average y translation (mm)	-0.15
Average z translation (mm)	-0.08
Average x rotation (deg)	0.41
Average y rotation (deg)	0.23
Average z rotation (deg)	-0.07

Outliers

Total outliers (%)	0.75
Outliers (b=500 s/mm^2)	9.26
Outliers (b=1000 s/mm^2)	0.41
Outliers (b=2000 s/mm^2)	0.22
Outliers (PE dir=[0. 1. 0.])	0.68
Outliers (PE dir=[0. -1. 0.])	0.53

SNR/CNR

Average SNR (b=0 s/mm^2)	32.26
Average CNR (b=500 s/mm^2)	1.71
Average CNR (b=1000 s/mm^2)	2.80
Average CNR (b=2000 s/mm^2)	2.27

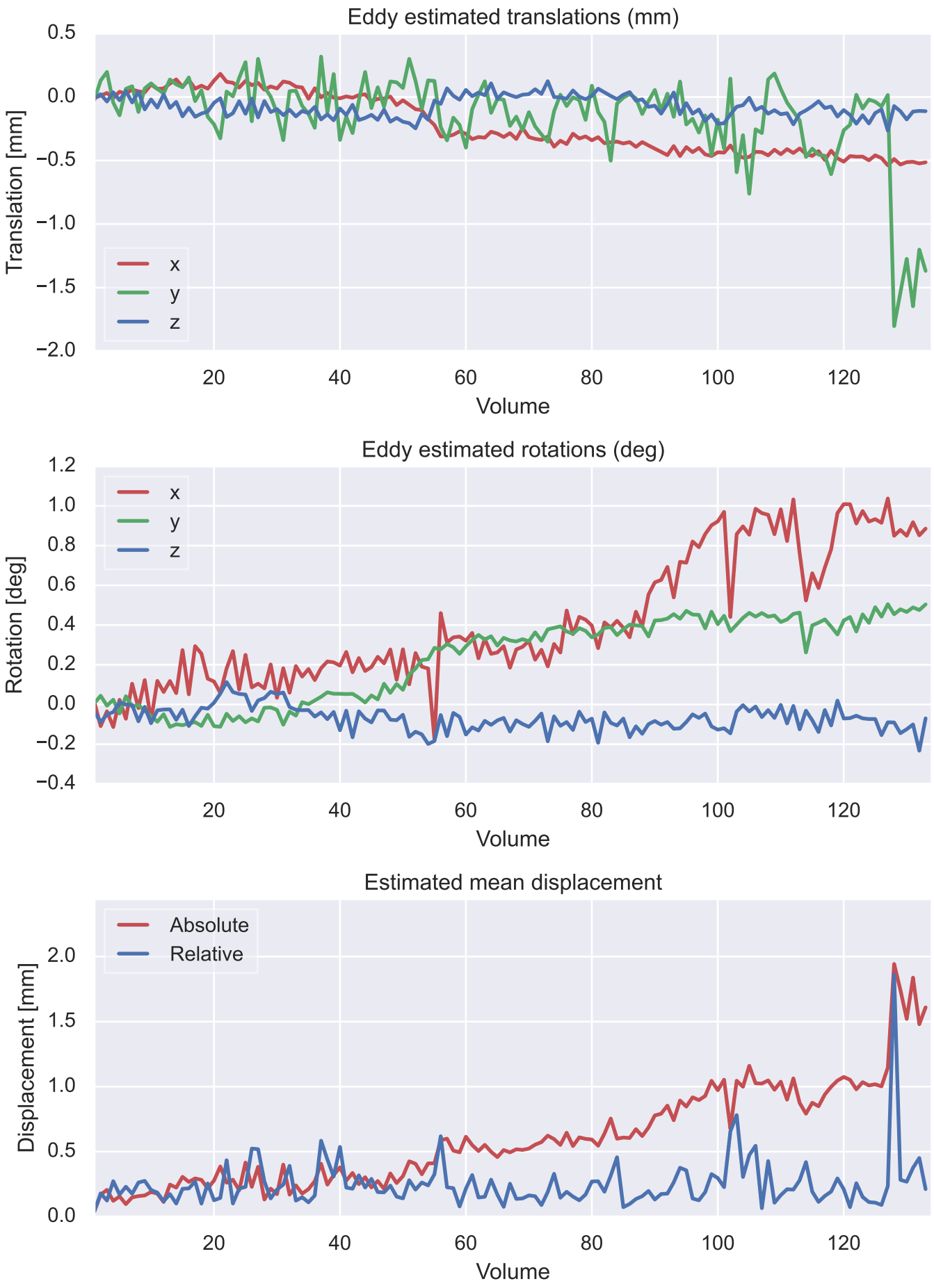
Eddy currents

Std Dev EC linear term (x)	0.02
Std Dev EC linear term (y)	0.05
Std Dev EC linear term (z)	0.05

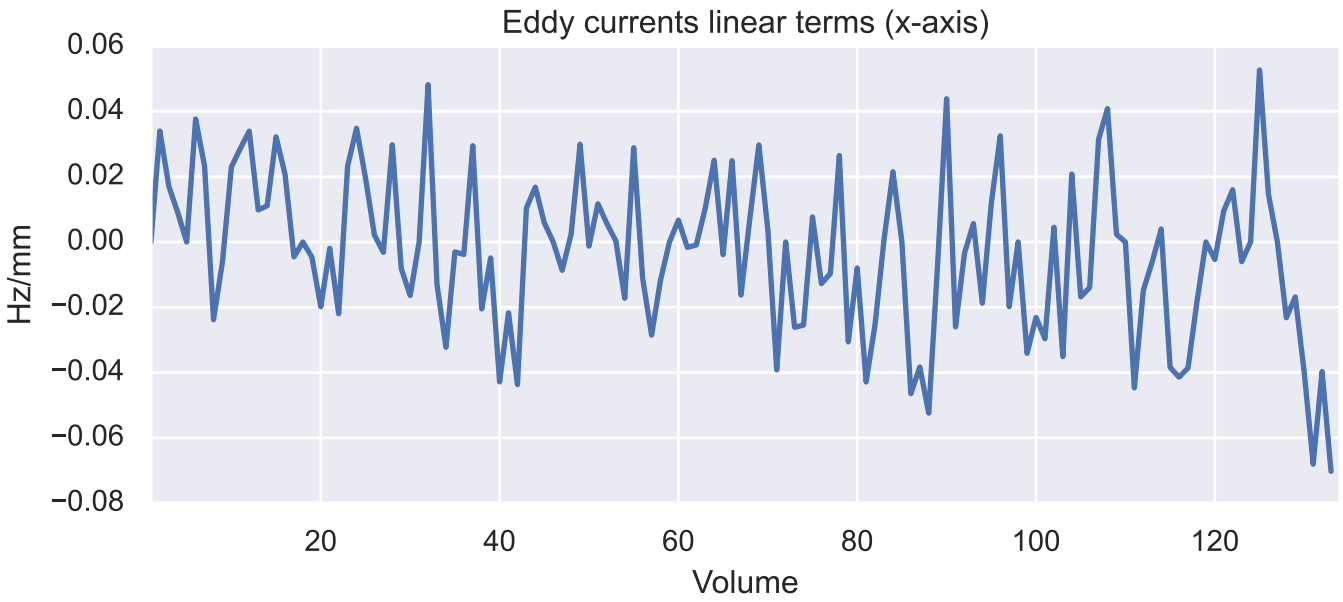
Susceptibility distortions

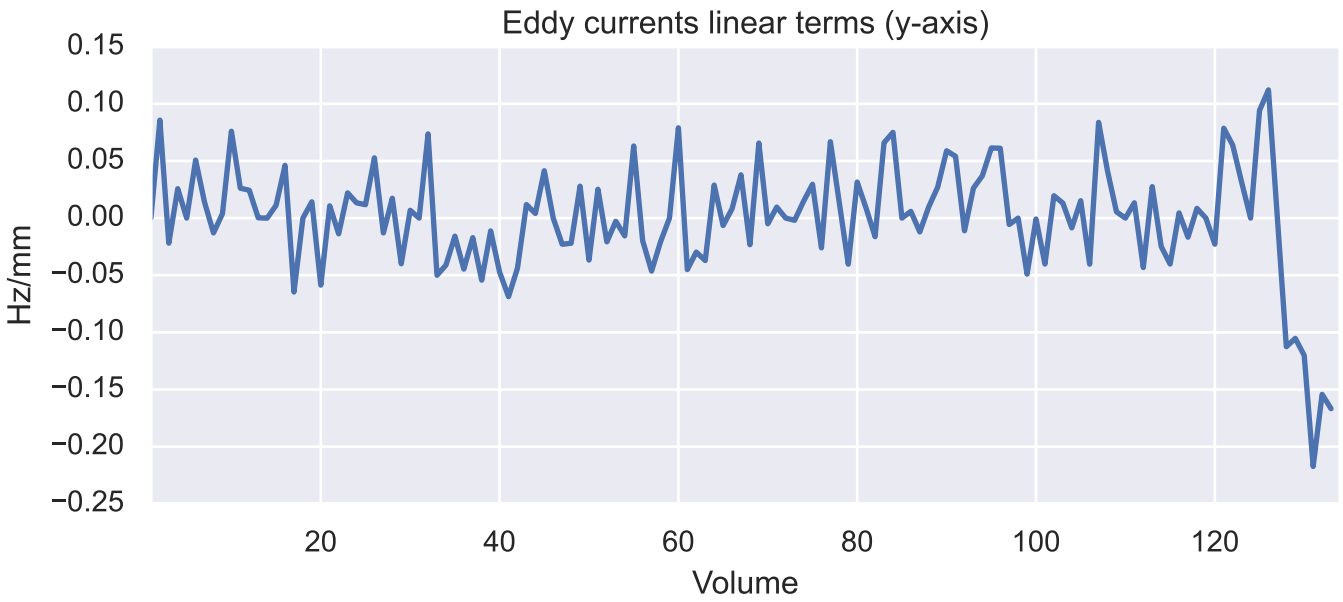
Std Dev voxel displacement	1.67

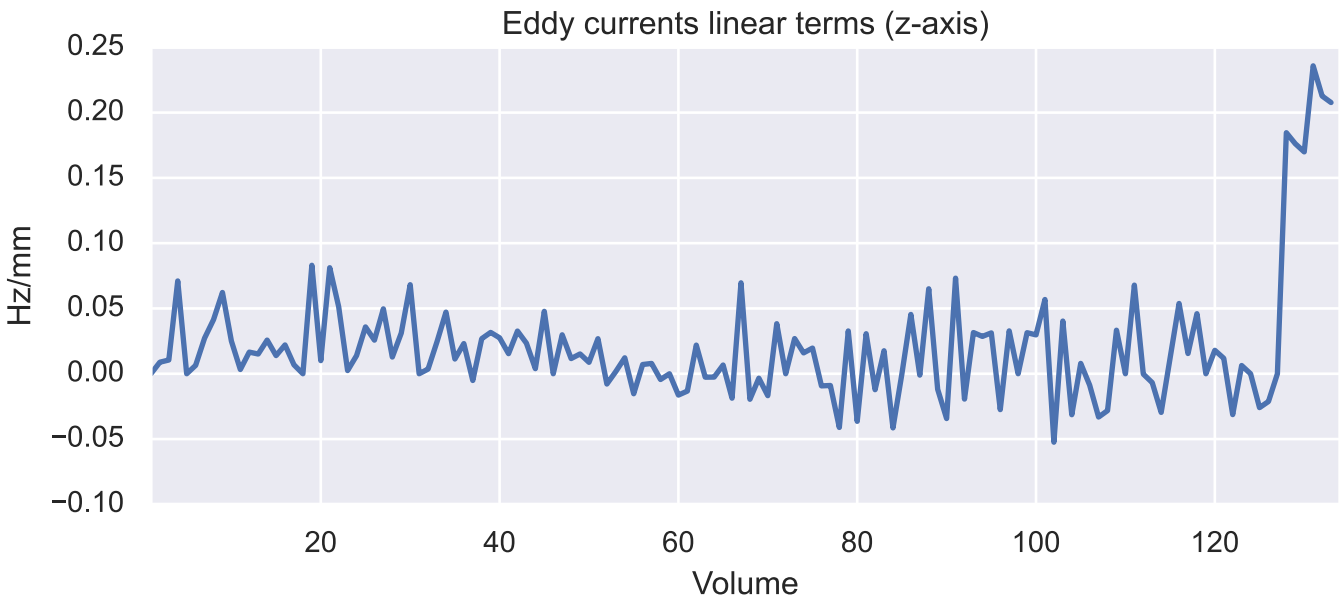
Subject 07_eddy_outputs/dwi_eddy.nii.gz



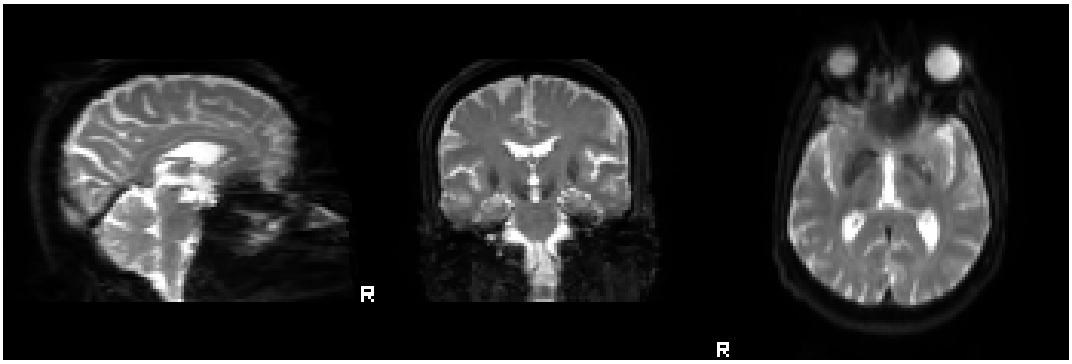
Subject 07_eddy_outputs/dwi_eddy.nii.gz

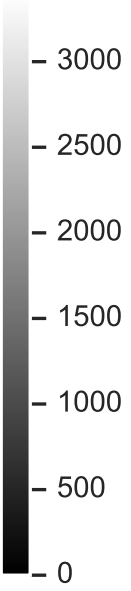




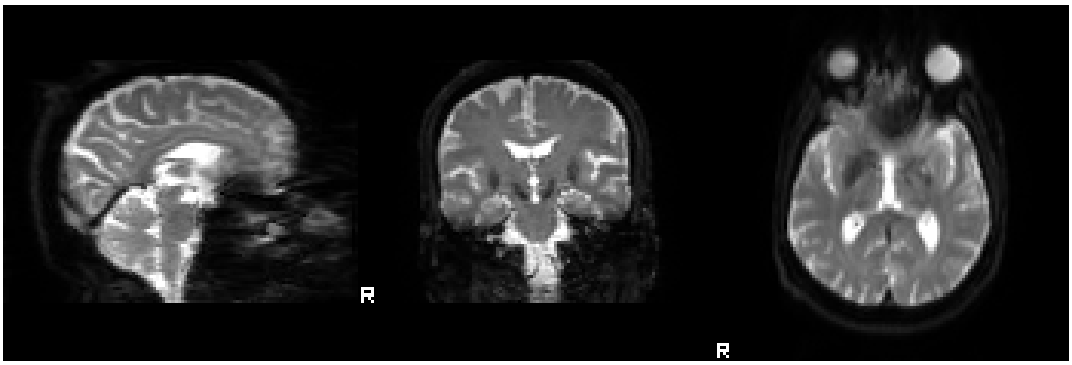


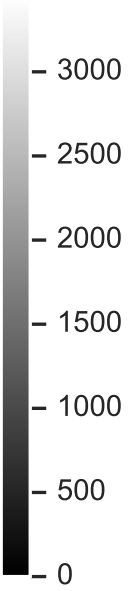
Average b0 signal (PE=[[0. 1. 0.]])



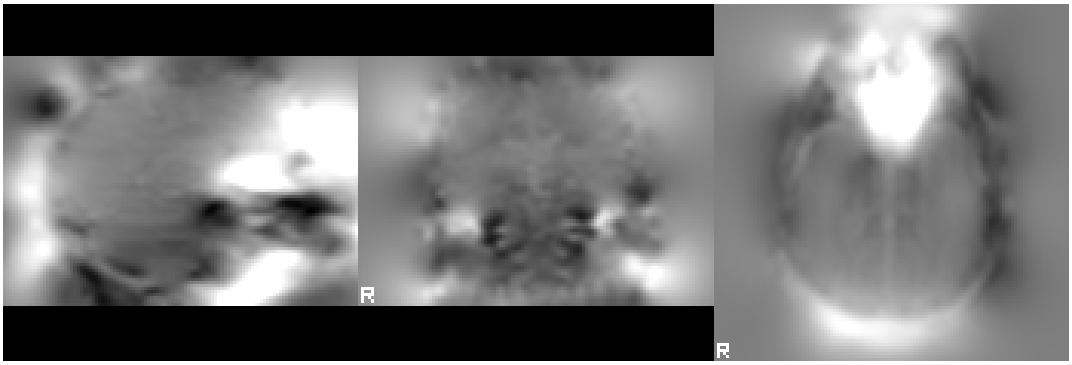


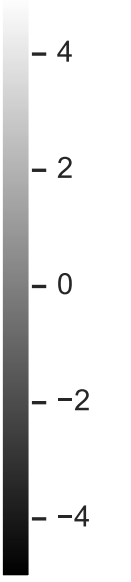
Average b0 signal (PE=[[0. -1. 0.]])



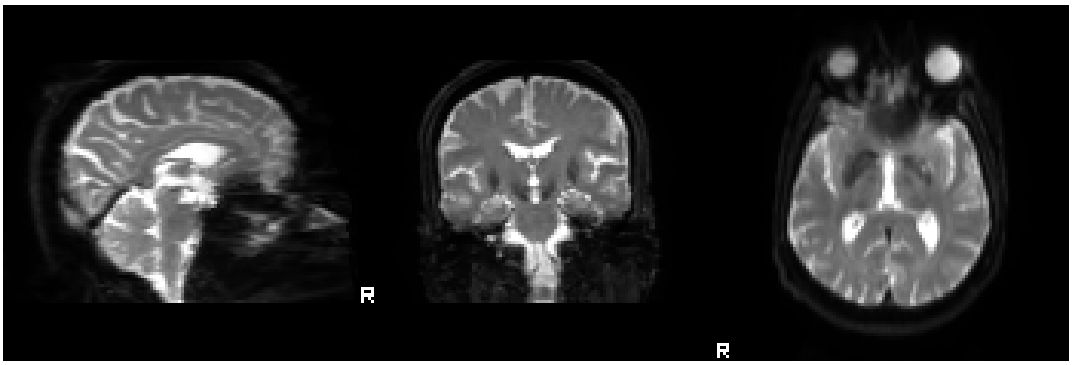


Voxel displacement map



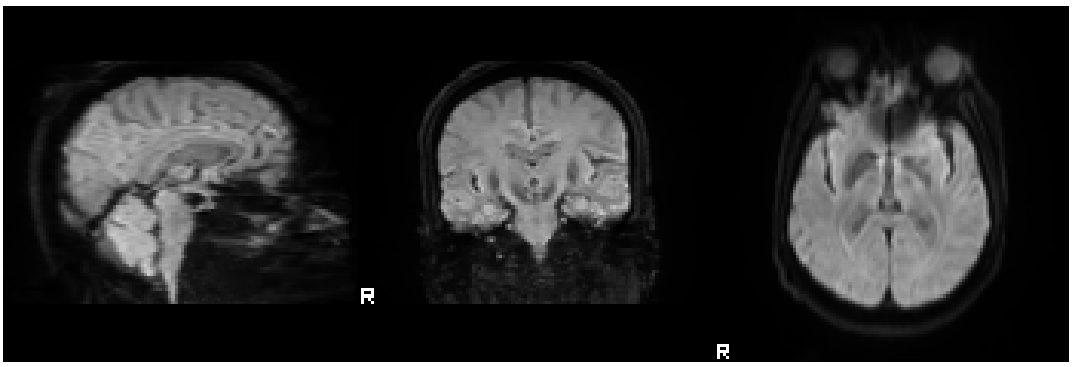


Average DW signal (b=0)



3000
2500
2000
1500
1000
500
0

Average DW signal (b=500)



1600
1400
1200
1000
800
600
400
200
0

Average DW signal (b=1000)

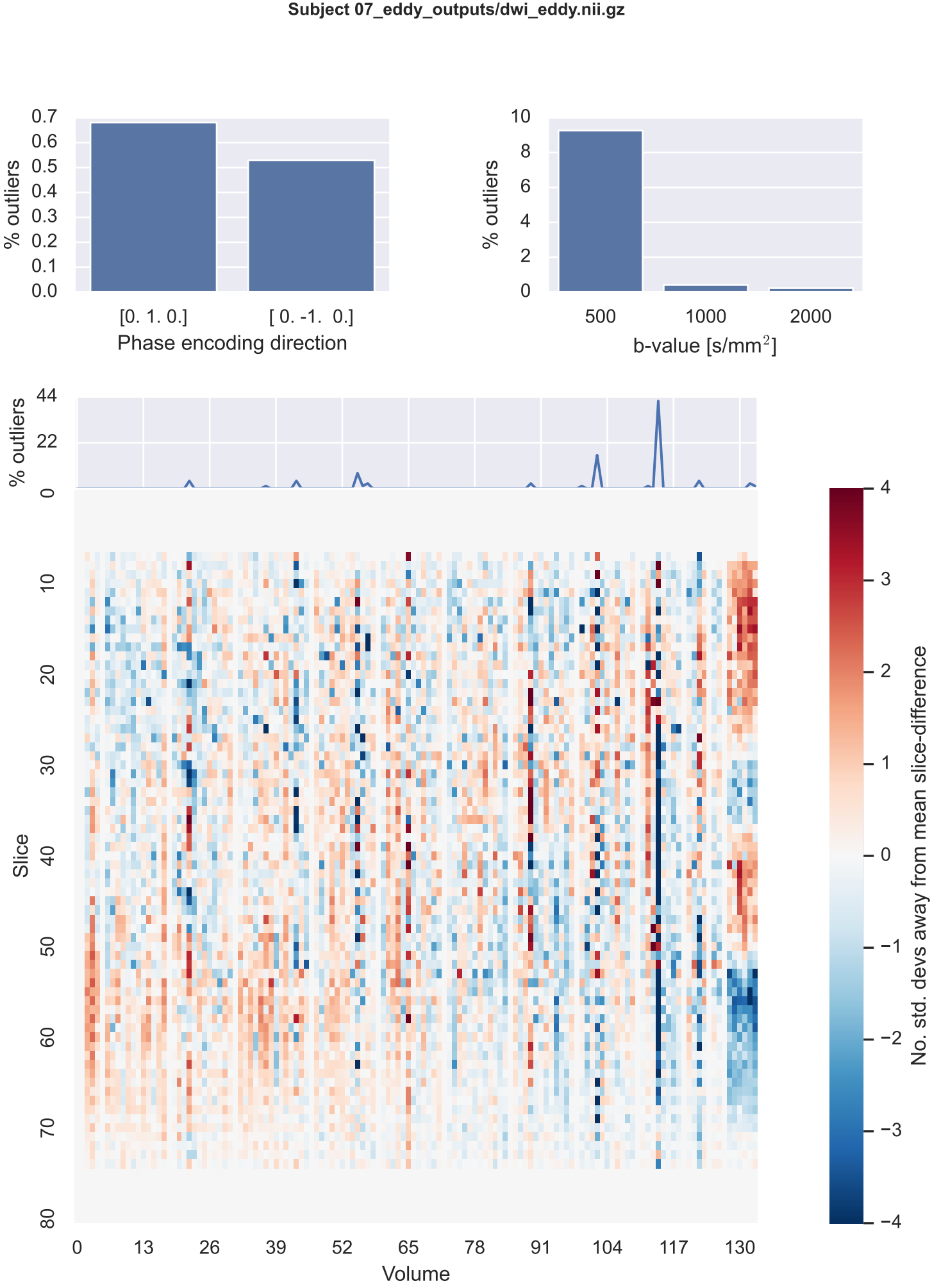


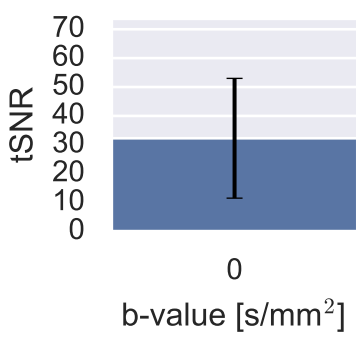
1200
1000
800
600
400
200
0

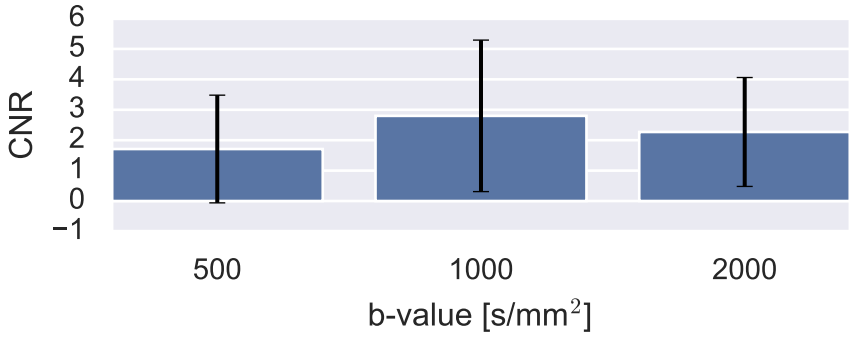
Average DW signal (b=2000)



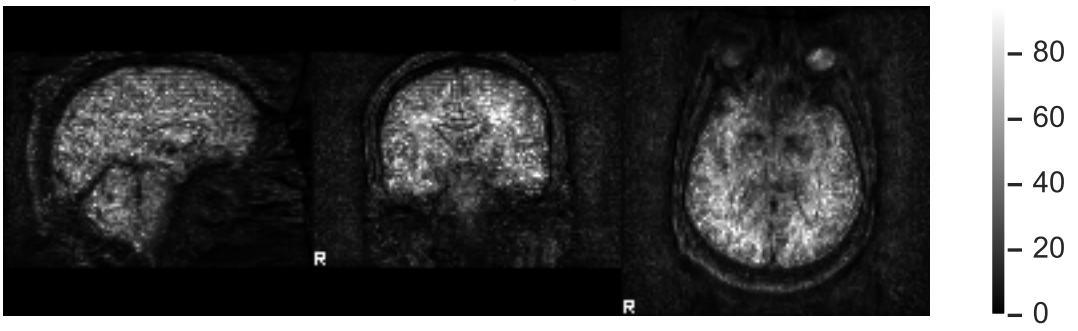
700
600
500
400
300
200
100
0


Subject 07_eddy_outputs/dwi_eddy.nii.gz
% outliers
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
[0. 1. 0.]
[0. -1. 0.]
Phase encoding direction
% outliers
10
8
6
4
2
0
500
1000
2000
b-value [s/mm^2]
% outliers
44
22
0
Slice
10
20
30
40
50
60
70
80
0
13
26
39
52
65
78
91
104
117
130
Volume
4
3
2
1
0
−1
−2
−3
−4
No. std. devs away from mean slice-difference

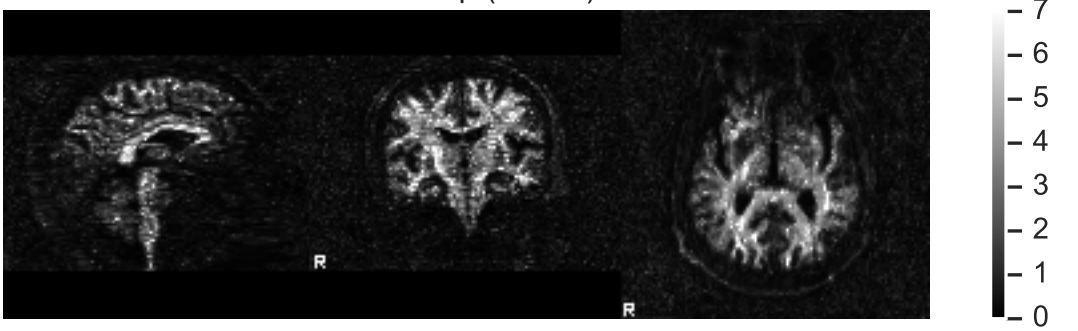




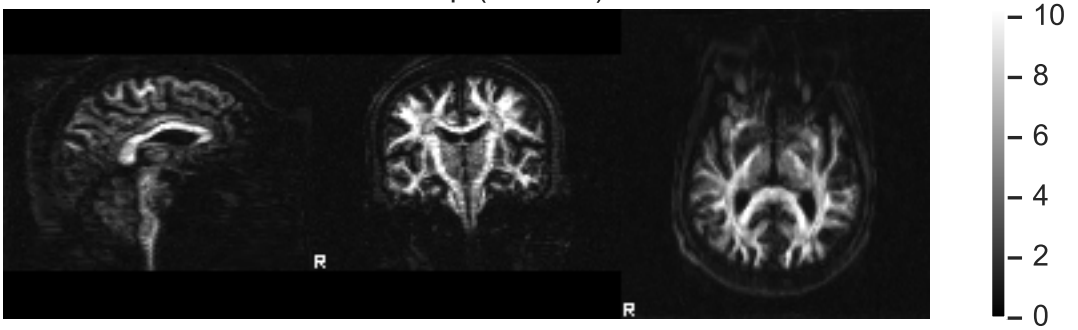
tSNR map (b=0)



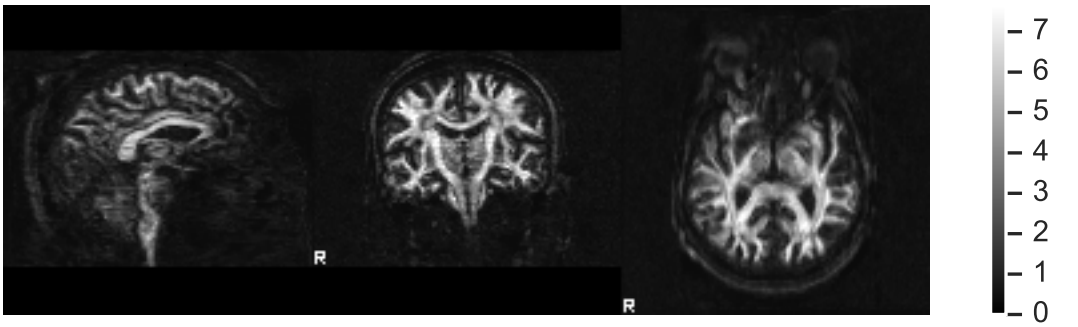
CNR map (b=500)

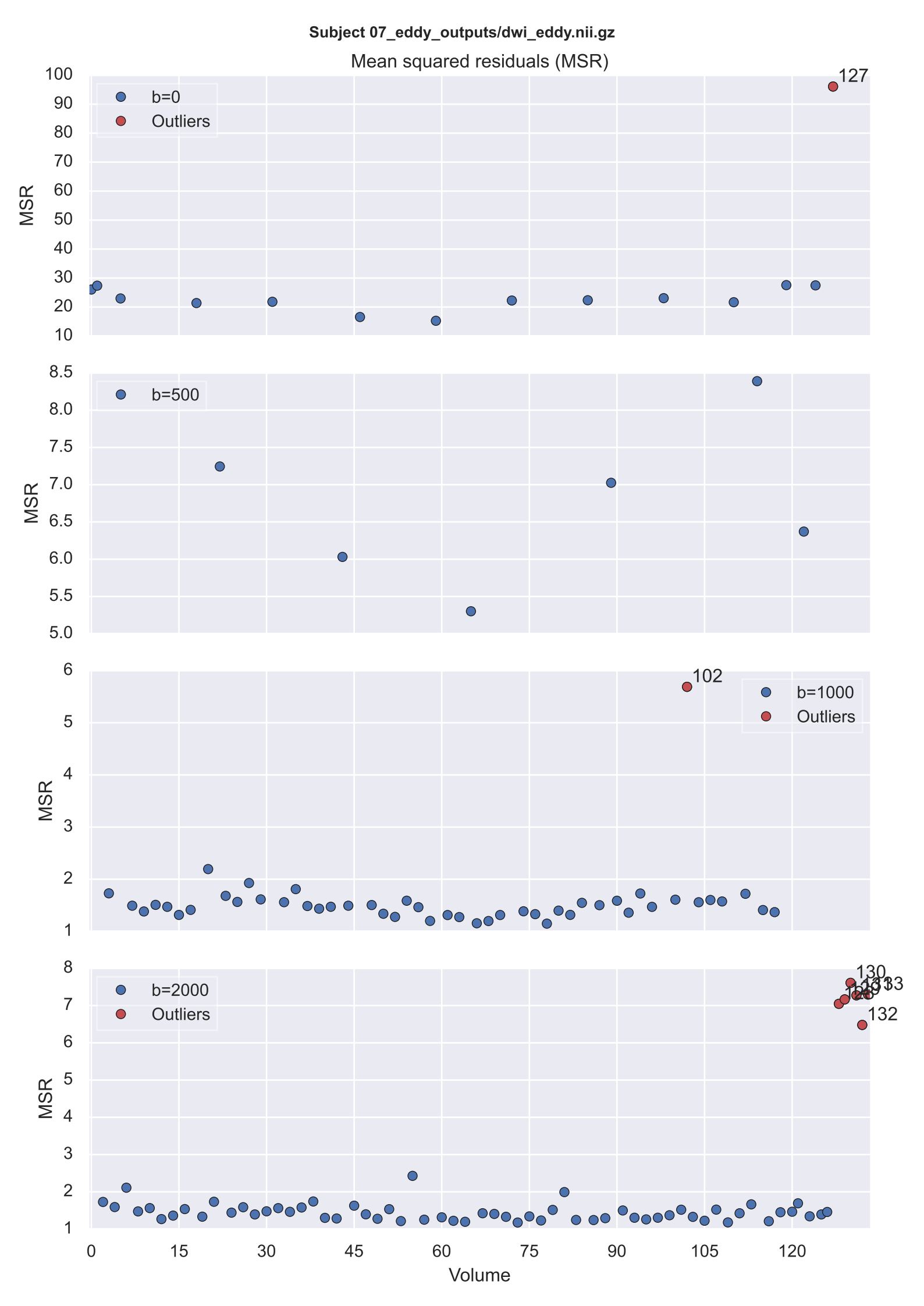


CNR map (b=1000)



CNR map (b=2000)




Subject 07_eddy_outputs/dwi_eddy.nii.gz
Mean squared residuals (MSR)
b=0
Outliers
127
MSR
b=500
MSR
102
b=1000
Outliers
MSR
b=2000
Outliers
130
132
MSR
Volume